

PROJECTPROPOSAL

PROJECTPILOT AI DISCOVERY ENGINE

Scope Architecture & Estimate: software house

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Business Domain:	EDUCATION
Target Platforms:	MOBILE
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1. Executive Summary

The proposed project 'software house ' is designed to serve as a high-quality education platform. Built for mobile, it incorporates core features including auth, notifications, admin, search. The proposed EdTech platform delivers a virtual learning environment for students and teachers. It emphasizes course management, content consumption, and interactive quizzes. The solution is optimized for varying network conditions, allowing students to access videos and text materials with minimum buffer times.

2. Project Objectives

The primary business objective is to architect and launch a scalable, secure, and production-quality software application. Through ProjectPilot AI's automated discovery session, we have analyzed requirements, selected platforms, and structured business modules to maximize engineering velocity while keeping operational expenditures low.

■ Time-to-Market	Accelerate launch cycles by using a modular, containerized stack (FastAPI & React).
■ Compliance & Security	Integrate native encryption, OAuth2 verification, and domain security mapped to education standards.
■ Future Expansion	Implement clean API separation, allowing effortless third-party integrations and smart automation additions.

3. Scope of Work & Features

The total scope covers a customized combination of standard features selected by the client, combined with intelligent additions recommended by the AI Architect based on industry standards.

Core Selected Features

- ✓ Auth

✓ Admin

✓ Dashboard
- ✓ Notifications

✓ Search

AI-Recommended Enhancements

Recommended Feature	Value-Add & Purpose	Effort
Advanced Analytics Module	Custom reporting and visual data monitoring widgets.	Medium

4. Solution Architecture

The platform is structured on a modern, decoupled architecture. Client interactions happen via a responsive Single Page Application (SPA), sending secure, rate-limited requests to a high-speed async backend.

Presentation Layer	Service Layer	Persistence Layer
Vite + React SPA Framer Motion UI Responsive Design	FastAPI Async server Uvicorn Router JSON Serialization	PostgreSQL relational DB SQLAlchemy Engine Redis Cache

Technology Stack Specifications

Infrastructure Layer	Recommended Technology Stack
Frontend	React Native (Expo, NativeWind) for mobile app deployment
Backend	FastAPI (Python), WebSockets for real-time quiz updates and chat rooms.
Database	MongoDB or PostgreSQL with JSON fields for course curricula.
Hosting & Infrastructure	Cloudflare CDN for video caching, AWS S3 for media storage, AWS Amplify.
External Integrations	Zoom API for live lectures, Vimeo API for video hosting, Zoom SDK for inside-app calls.

5. Project Implementation Roadmap

Phase / Milestone	Deliverables Scope	Est. Duration
Phase 1: Foundations & User Core	- Database Schema setup - Core API configurations - User Accounts & Authentication module	2 Weeks
Phase 2: Product Core Modules	- Notifications module development - Admin module development - Search module development - Dashboard module development	3 Weeks
Phase 3: QA, Security & Deployment	- Quality Assurance & Integration tests - Performance optimization runs - Cloud production environment deployment	1 Week

Agile Sprint Planning

Sprint	Sprint Objectives & Effort	Deliverables Checklist
Sprint 1	<i>Objectives:</i> Establish infrastructure, build base APIs, and set up landing dashboard frames. Effort: 125 Hours	- Docker infrastructure and API templates initialized - Database migrations complete and connected locally - User account registration UI & Backend logic verified
Sprint 2	<i>Objectives:</i> Complete end-to-end integration tests, resolve UI issues, and deploy to staging. Effort: 125 Hours	- Full integration unit test suite checks passing - Responsive styling issues resolved across web/mobile viewports - Continuous Integration (CI/CD) pipelines running successfully

6. Team Allocation & Budget Breakdown

Based on project parameters, the scope of selected features, and platform requirements, we have computed target effort metrics and engineered a precise allocation layout.

Total Est. Effort	Est. Timeline Duration	Engineering Team Size
250 hours	4-6 weeks	2 resources

Allocated Roles: Frontend Developer, Backend Developer

Category Component	Scope Details	Estimated Cost (USD)
UI/UX Design Phase	User experience wireframes, interactive mockup builds	\$3,750.00
Frontend Development	React / Mobile application interface construction	\$8,750.00
Backend Engineering	FastAPI endpoints, ORM connections, databases setup	\$7,500.00
Quality Assurance	End-to-end integration tests, responsiveness check	\$3,000.00
Project Management	Agile coordination, milestones, developer logs tracking	\$2,000.00
Hosting & Cloud Infrastructure	Initial VPS staging sandbox setup & pipeline actions	\$500.00
TOTAL BUDGET	Estimated total investment, excluding hosting	\$25,000.00

7. Technical Risks & Mitigations

Identified Risk Factor	Proposed Mitigation Strategy
Video streaming buffering on low bandwidth connections	Implement adaptive bitrate streaming (HLS/DASH) and utilize Cloudflare CDN.
Cheating on quizzes and assignments	Randomize question orders, disable copy-paste actions on quiz pages, and add strict submission timeouts.
Media storage costs rising quickly with course uploads	Apply automatic video compression and transcoding tools (AWS Elemental MediaConvert) on upload.

8. Project Execution Next Steps

To initiate the development workflow, we recommend the following phases:

- 1. Discovery Alignment:** Conduct a developer kick-off session to align architectural preferences and integrations.
- 2. Environments Initialization:** Provision SQLite/PostgreSQL staging databases and hook continuous integration (CI) actions.
- 3. Milestone Sprint 1:** Execute Sprint 1 scope, starting with authentication configurations and dashboard scaffolding.